

Homeostasis

Some things about human activity are easy to observe. When you run or swim or write the answer to a test question, you can see your body at work. But behind the scenes, your body works constantly to do something that is difficult to see—maintaining a controlled, stable internal environment. This process is called **homeostasis**, which means “keeping things the same.” *Homeostasis describes the relatively constant internal conditions that organisms maintain despite changes in internal and external environments.* Homeostasis may not be obvious, but for a living organism, it’s a matter of life or death.

Feedback Inhibition If you’ve ever watched someone driving a car down a straight road, you may have noticed the person slightly moving the wheel left and right, adjusting direction to keep the vehicle in the middle of the lane. In a sense, that’s how the systems of the body work, keeping internal conditions within a certain range, never allowing them to go too far one way or the other.

A Nonliving Example One way to understand homeostasis is to look at a nonliving system that keeps conditions within a certain range, such as a home heating system. In many homes, heat is supplied by a furnace. When the temperature within the house drops below a set point, a thermostat sensor switches the furnace on. Heat produced by the furnace warms the house. When the temperature rises above the set point, the thermostat switches the furnace off, keeping the temperature within a narrow range.

A system like this example is controlled by feedback inhibition. **Feedback inhibition**, or negative feedback, is the process in which a stimulus produces a response that opposes the original stimulus. **Figure 27-3** summarizes the feedback inhibition process in a home heating system. When the furnace is switched on, it produces a product (heat) that changes the environment of the house by raising the air temperature. This environmental change then “feeds back” to “inhibit” the operation of the furnace. In other words, heat from the furnace eventually raises the temperature high enough to trigger a feedback signal that switches the furnace off. Systems controlled by feedback inhibition are generally very stable.

READING CHECK **Apply Concepts** Describe another example of a nonliving system that requires constant adjustment.

BUILD VOCABULARY

Root Words The root word *stasis* means “a state of balance or equilibrium.”



VIDEO

Learn how feedback inhibition is used to return the body to its normal, homeostatic state.

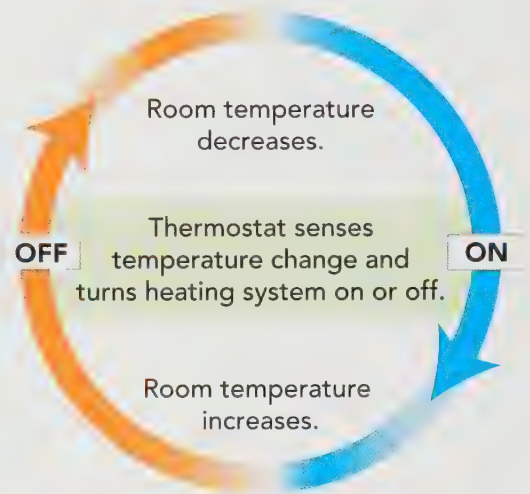


Figure 27-3
Feedback Inhibition

A home heating system uses a feedback loop to maintain a stable, comfortable environment within a house.